

Class 9

Stephanie Solis (PID A18572681)

Importing Candy Data

```
candy_file <- "candy-data.csv"
candy <- read.csv(candy_file, row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>  
filter(row.names(candy)=="Twix") |>  
select(winpercent)
```

```
      winpercent  
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy["Snickers", ]$winpercent
```

```
[1] 76.67378
```

Q4. What is the winpercent value for "Kit Kat"?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

```
library("skimr")  
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Winpercent

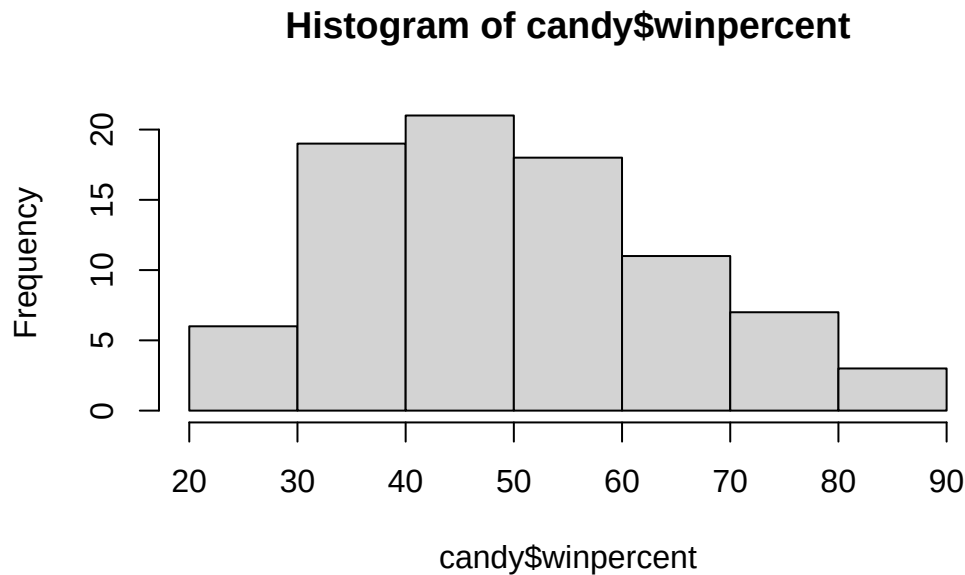
Q7. What do you think a zero and one represent for the candy\$chocolate column?

A zero indicates that it does not contain chocolate, while a one indicates that it does contain chocolate.

Exploratory Analysis

Q8. Plot a histogram of winpercent values

```
hist(candy$winpercent)
```



Q9. Is the distribution of winpercent values symmetrical?

No, they are not symmetrical.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

Above 50%

Q11. On average is chocolate candy higher or lower ranked than fruit candy? 1. Find all chocolate candy 2. Get their winpercent values 3. Find the mean 4. Find all fruit candy 5. Get their winpercent values 6. Find the mean 7. Compare the two means

```
choc.candy <- candy[candy$chocolate == 1, ]  
choc.win <- choc.candy$winpercent  
mean(choc.win)
```

```
[1] 60.92153
```

```
fruit.win <- candy[candy$fruity == 1,]$winpercent  
mean(fruit.win)
```

```
[1] 44.11974
```

Q12. Is this difference statistically significant?

```
t.test(choc.win, fruit.win)
```

Welch Two Sample t-test

```
data:  choc.win and fruit.win  
t = 6.2582, df = 68.882, p-value = 2.871e-08  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
 11.44563 22.15795  
sample estimates:  
mean of x mean of y  
 60.92153  44.11974
```

There is a statistical difference.

Q13. What are the five least liked candy types in this set?

```
y <- c("y", "a", "z")
sort(y)
```

```
[1] "a" "y" "z"
```

```
y
```

```
[1] "y" "a" "z"
```

```
order(y)
```

```
[1] 2 1 3
```

```
ord.ind <- order(candy$winpercent)
candy[ord.ind,]
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0
Root Beer Barrels	0	0	0	0	0
Sugar Daddy	0	0	1	0	0
One dime	0	0	0	0	0
Sugar Babies	0	0	1	0	0
Haribo Happy Cola	0	0	0	0	0
Caramel Apple Pops	0	1	1	0	0
Strawberry bon bons	0	1	0	0	0
Sixlets	1	0	0	0	0
Ring pop	0	1	0	0	0
Chewy Lemonhead Fruit Mix	0	1	0	0	0
Red vines	0	1	0	0	0
Pixie Sticks	0	0	0	0	0
Nestle Smarties	1	0	0	0	0
Candy Corn	0	0	0	0	0

Charleston Chew	1	0	0	0	1
Warheads	0	1	0	0	0
Lemonhead	0	1	0	0	0
Fun Dip	0	1	0	0	0
Now & Later	0	1	0	0	0
Dum Dums	0	1	0	0	0
Pop Rocks	0	1	0	0	0
Laffy Taffy	0	1	0	0	0
Werther's Original Caramel	0	0	1	0	0
Haribo Twin Snakes	0	1	0	0	0
Dots	0	1	0	0	0
Runts	0	1	0	0	0
Tootsie Roll Juniors	1	0	0	0	0
Fruit Chews	0	1	0	0	0
Welch's Fruit Snacks	0	1	0	0	0
Twizzlers	0	1	0	0	0
Tootsie Roll Midgies	1	0	0	0	0
Smarties candy	0	1	0	0	0
One quarter	0	0	0	0	0
Payday	0	0	0	1	1
Mike & Ike	0	1	0	0	0
Gobstopper	0	1	0	0	0
Trolli Sour Bites	0	1	0	0	0
Mounds	1	0	0	0	0
Tootsie Pop	1	1	0	0	0
Whoppers	1	0	0	0	0
Tootsie Roll Snack Bars	1	0	0	0	0
Almond Joy	1	0	0	1	0
Haribo Sour Bears	0	1	0	0	0
Air Heads	0	1	0	0	0
Sour Patch Tricksters	0	1	0	0	0
Lifesavers big ring gummies	0	1	0	0	0
Mr Good Bar	1	0	0	1	0
Swedish Fish	0	1	0	0	0
Milk Duds	1	0	1	0	0
Skittles wildberry	0	1	0	0	0
Nerds	0	1	0	0	0
Hershey's Kisses	1	0	0	0	0
Hershey's Milk Chocolate	1	0	0	0	0
Baby Ruth	1	0	1	1	1
Haribo Gold Bears	0	1	0	0	0
Junior Mints	1	0	0	0	0
Hershey's Special Dark	1	0	0	0	0

Snickers Crisper	1	0	1	1	0
Sour Patch Kids	0	1	0	0	0
Milky Way Midnight	1	0	1	0	1
Hershey's Krackel	1	0	0	0	0
Skittles original	0	1	0	0	0
Milky Way Simply Caramel	1	0	1	0	0
Rolo	1	0	1	0	0
Nestle Crunch	1	0	0	0	0
M&M's	1	0	0	0	0
100 Grand	1	0	1	0	0
Starburst	0	1	0	0	0
3 Musketeers	1	0	0	0	1
Peanut M&Ms	1	0	0	1	0
Nestle Butterfinger	1	0	0	1	0
Peanut butter M&M's	1	0	0	1	0
Reese's stuffed with pieces	1	0	0	1	0
Milky Way	1	0	1	0	1
Reese's pieces	1	0	0	1	0
Snickers	1	0	1	1	1
Kit Kat	1	0	0	0	0
Twix	1	0	1	0	0
Reese's Miniatures	1	0	0	1	0
Reese's Peanut Butter cup	1	0	0	1	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Nik L Nip		0	0	0		1		0.197
Boston Baked Beans		0	0	0		1		0.313
Chiclets		0	0	0		1		0.046
Super Bubble		0	0	0		0		0.162
Jawbusters		0	1	0		1		0.093
Root Beer Barrels		0	1	0		1		0.732
Sugar Daddy		0	0	0		0		0.418
One dime		0	0	0		0		0.011
Sugar Babies		0	0	0		1		0.965
Haribo Happy Cola		0	0	0		1		0.465
Caramel Apple Pops		0	0	0		0		0.604
Strawberry bon bons		0	1	0		1		0.569
Sixlets		0	0	0		1		0.220
Ring pop		0	1	0		0		0.732
Chewy Lemonhead Fruit Mix		0	0	0		1		0.732
Red vines		0	0	0		1		0.581
Pixie Sticks		0	0	0		1		0.093
Nestle Smarties		0	0	0		1		0.267
Candy Corn		0	0	0		1		0.906

Charleston Chew	0	0	1	0	0.604
Warheads	0	1	0	0	0.093
Lemonhead	0	1	0	0	0.046
Fun Dip	0	1	0	0	0.732
Now & Later	0	0	0	1	0.220
Dum Dums	0	1	0	0	0.732
Pop Rocks	0	1	0	1	0.604
Laffy Taffy	0	0	0	0	0.220
Werther's Original Caramel	0	1	0	0	0.186
Haribo Twin Snakes	0	0	0	1	0.465
Dots	0	0	0	1	0.732
Runts	0	1	0	1	0.872
Tootsie Roll Juniors	0	0	0	0	0.313
Fruit Chews	0	0	0	1	0.127
Welch's Fruit Snacks	0	0	0	1	0.313
Twizzlers	0	0	0	0	0.220
Tootsie Roll Midgies	0	0	0	1	0.174
Smarties candy	0	1	0	1	0.267
One quarter	0	0	0	0	0.011
Payday	0	0	1	0	0.465
Mike & Ike	0	0	0	1	0.872
Gobstopper	0	1	0	1	0.906
Trolli Sour Bites	0	0	0	1	0.313
Mounds	0	0	1	0	0.313
Tootsie Pop	0	1	0	0	0.604
Whoppers	1	0	0	1	0.872
Tootsie Roll Snack Bars	0	0	1	0	0.465
Almond Joy	0	0	1	0	0.465
Haribo Sour Bears	0	0	0	1	0.465
Air Heads	0	0	0	0	0.906
Sour Patch Tricksters	0	0	0	1	0.069
Lifesavers big ring gummies	0	0	0	0	0.267
Mr Good Bar	0	0	1	0	0.313
Swedish Fish	0	0	0	1	0.604
Milk Duds	0	0	0	1	0.302
Skittles wildberry	0	0	0	1	0.941
Nerds	0	1	0	1	0.848
Hershey's Kisses	0	0	0	1	0.127
Hershey's Milk Chocolate	0	0	1	0	0.430
Baby Ruth	0	0	1	0	0.604
Haribo Gold Bears	0	0	0	1	0.465
Junior Mints	0	0	0	1	0.197
Hershey's Special Dark	0	0	1	0	0.430

Snickers Crisper	1	0	1	0	0.604
Sour Patch Kids	0	0	0	1	0.069
Milky Way Midnight	0	0	1	0	0.732
Hershey's Krackel	1	0	1	0	0.430
Skittles original	0	0	0	1	0.941
Milky Way Simply Caramel	0	0	1	0	0.965
Rolo	0	0	0	1	0.860
Nestle Crunch	1	0	1	0	0.313
M&M's	0	0	0	1	0.825
100 Grand	1	0	1	0	0.732
Starburst	0	0	0	1	0.151
3 Musketeers	0	0	1	0	0.604
Peanut M&Ms	0	0	0	1	0.593
Nestle Butterfinger	0	0	1	0	0.604
Peanut butter M&M's	0	0	0	1	0.825
Reese's stuffed with pieces	0	0	0	0	0.988
Milky Way	0	0	1	0	0.604
Reese's pieces	0	0	0	1	0.406
Snickers	0	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Twix	1	0	1	0	0.546
Reese's Miniatures	0	0	0	0	0.034
Reese's Peanut Butter cup	0	0	0	0	0.720

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Boston Baked Beans	0.511	23.41782
Chiclets	0.325	24.52499
Super Bubble	0.116	27.30386
Jawbusters	0.511	28.12744
Root Beer Barrels	0.069	29.70369
Sugar Daddy	0.325	32.23100
One dime	0.116	32.26109
Sugar Babies	0.767	33.43755
Haribo Happy Cola	0.465	34.15896
Caramel Apple Pops	0.325	34.51768
Strawberry bon bons	0.058	34.57899
Sixlets	0.081	34.72200
Ring pop	0.965	35.29076
Chewey Lemonhead Fruit Mix	0.511	36.01763
Red vines	0.116	37.34852
Pixie Sticks	0.023	37.72234
Nestle Smarties	0.976	37.88719
Candy Corn	0.325	38.01096

Charleston Chew	0.511	38.97504
Warheads	0.116	39.01190
Lemonhead	0.104	39.14106
Fun Dip	0.325	39.18550
Now & Later	0.325	39.44680
Dum Dums	0.034	39.46056
Pop Rocks	0.837	41.26551
Laffy Taffy	0.116	41.38956
Werther's Original Caramel	0.267	41.90431
Haribo Twin Snakes	0.465	42.17877
Dots	0.511	42.27208
Runts	0.279	42.84914
Tootsie Roll Juniors	0.511	43.06890
Fruit Chews	0.034	43.08892
Welch's Fruit Snacks	0.313	44.37552
Twizzlers	0.116	45.46628
Tootsie Roll Midgies	0.011	45.73675
Smarties candy	0.116	45.99583
One quarter	0.511	46.11650
Payday	0.767	46.29660
Mike & Ike	0.325	46.41172
Gobstopper	0.453	46.78335
Trolli Sour Bites	0.255	47.17323
Mounds	0.860	47.82975
Tootsie Pop	0.325	48.98265
Whoppers	0.848	49.52411
Tootsie Roll Snack Bars	0.325	49.65350
Almond Joy	0.767	50.34755
Haribo Sour Bears	0.465	51.41243
Air Heads	0.511	52.34146
Sour Patch Tricksters	0.116	52.82595
Lifesavers big ring gummies	0.279	52.91139
Mr Good Bar	0.918	54.52645
Swedish Fish	0.755	54.86111
Milk Duds	0.511	55.06407
Skittles wildberry	0.220	55.10370
Nerds	0.325	55.35405
Hershey's Kisses	0.093	55.37545
Hershey's Milk Chocolate	0.918	56.49050
Baby Ruth	0.767	56.91455
Haribo Gold Bears	0.465	57.11974
Junior Mints	0.511	57.21925
Hershey's Special Dark	0.918	59.23612

Snickers Crisper	0.651	59.52925
Sour Patch Kids	0.116	59.86400
Milky Way Midnight	0.441	60.80070
Hershey's Krackel	0.918	62.28448
Skittles original	0.220	63.08514
Milky Way Simply Caramel	0.860	64.35334
Rolo	0.860	65.71629
Nestle Crunch	0.767	66.47068
M&M's	0.651	66.57458
100 Grand	0.860	66.97173
Starburst	0.220	67.03763
3 Musketeers	0.511	67.60294
Peanut M&Ms	0.651	69.48379
Nestle Butterfinger	0.767	70.73564
Peanut butter M&M's	0.651	71.46505
Reese's stuffed with pieces	0.651	72.88790
Milky Way	0.651	73.09956
Reese's pieces	0.651	73.43499
Snickers	0.651	76.67378
Kit Kat	0.511	76.76860
Twix	0.906	81.64291
Reese's Miniatures	0.279	81.86626
Reese's Peanut Butter cup	0.651	84.18029

Q14. What are the top 5 all time favorite candy types out of this set?

```
head(candy[order(candy$winpercent, decreasing = TRUE), ])
```

	chocolate	fruity	caramel	peanut	almondy	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1
Reese's pieces	1	0	0		1	0
	crisped	rice	wafer	hard bar	pluribus	sugar
Reese's Peanut Butter cup		0	0	0	0	0.720
Reese's Miniatures		0	0	0	0	0.034
Twix		1	0	1	0	0.546
Kit Kat		1	0	1	0	0.313
Snickers		0	0	1	0	0.546
Reese's pieces		0	0	0	1	0.406

	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378
Reese's pieces	0.651	73.43499

Q15. Make a first barplot of candy ranking based on winpercent values.

We need a custom color vector

```
my_cols <- rep("black", nrow(candy))
my_cols[candy$chocolate==1] <- "chocolate"
my_cols[candy$fruity==1] <- "red"
my_cols
```

```
[1] "chocolate" "chocolate" "black"      "black"      "red"        "chocolate"
[7] "chocolate" "black"      "black"      "red"        "chocolate" "red"
[13] "red"        "red"        "red"        "red"        "red"        "red"
[19] "red"        "black"      "red"        "red"        "chocolate" "chocolate"
[25] "chocolate" "chocolate" "red"        "chocolate" "chocolate" "red"
[31] "red"        "red"        "chocolate" "chocolate" "red"        "chocolate"
[37] "chocolate" "chocolate" "chocolate" "chocolate" "chocolate" "red"
[43] "chocolate" "chocolate" "red"        "red"        "black"      "chocolate"
[49] "black"      "red"        "red"        "chocolate" "chocolate" "chocolate"
[55] "chocolate" "red"        "chocolate" "black"      "red"        "chocolate"
[61] "red"        "red"        "chocolate" "red"        "chocolate" "chocolate"
[67] "red"        "red"        "red"        "red"        "black"      "black"
[73] "red"        "red"        "red"        "chocolate" "chocolate" "chocolate"
[79] "red"        "chocolate" "red"        "red"        "red"        "black"
[85] "chocolate"
```

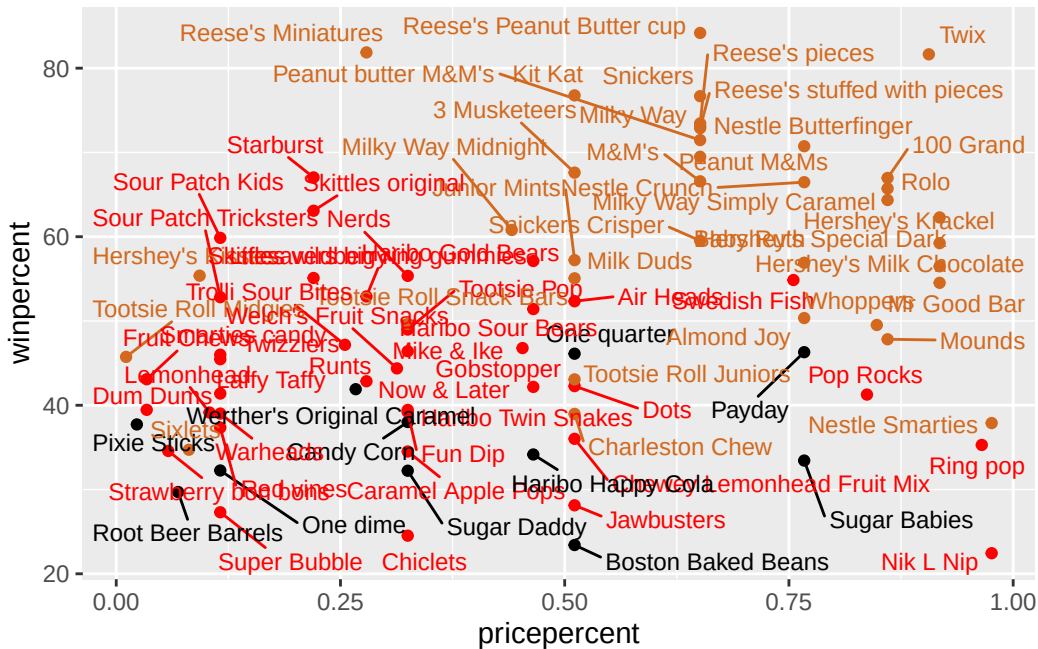
Taking a look at Pricepercent

```
library(ggrepel)
```

Loading required package: ggplot2

Loading required package: ggplot2

```
ggplot(candy) +
  aes(x=pricepercent, y=winpercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 50)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Peanut Butter Cup Miniatures.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
candy |> arrange(desc(pricepercent)) |> head(5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Nestle Smarties	1	0	0	0	0
Ring pop	0	1	0	0	0
Hershey's Krackel	1	0	0	0	0
Hershey's Milk Chocolate	1	0	0	0	0

	crisped	ricewafer	hard	bar	pluribus	sugarpercent
Nik L Nip	0	0	0		1	0.197
Nestle Smarties	0	0	0		1	0.267
Ring pop	0	1	0		0	0.732
Hershey's Krackel	1	0	1		0	0.430
Hershey's Milk Chocolate	0	0	1		0	0.430

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

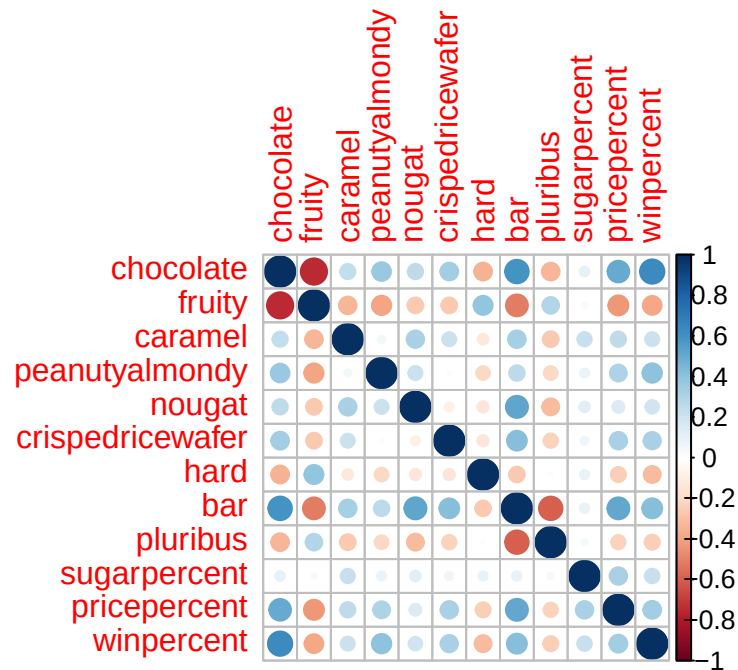
Exploring the correlation structure

```
cij <- cor(candy)
```

```
library(corrplot)
```

corrplot 0.95 loaded

```
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and fruity are anti-correlated.

Q23. Similarly, what two variables are most positively correlated?

Chocolate and bar are most positively correlated.

Principal Component Analysis (PCA)

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

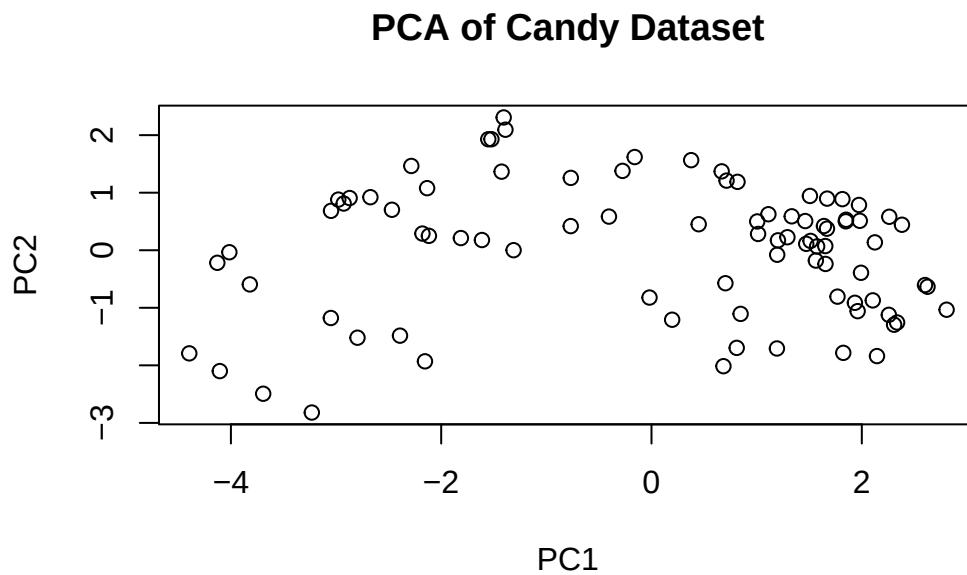
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317

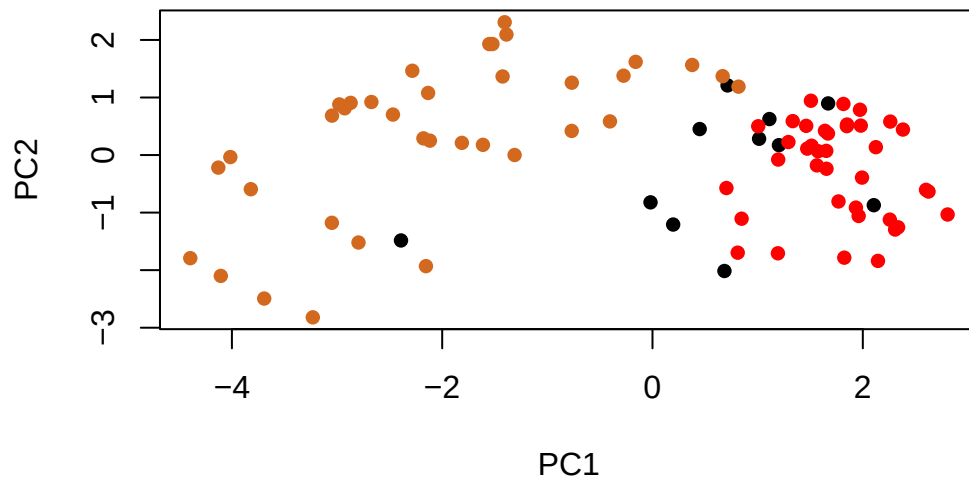
Cumulative Proportion 0.89998 0.93832 0.97071 0.98683 1.00000

Score plot...

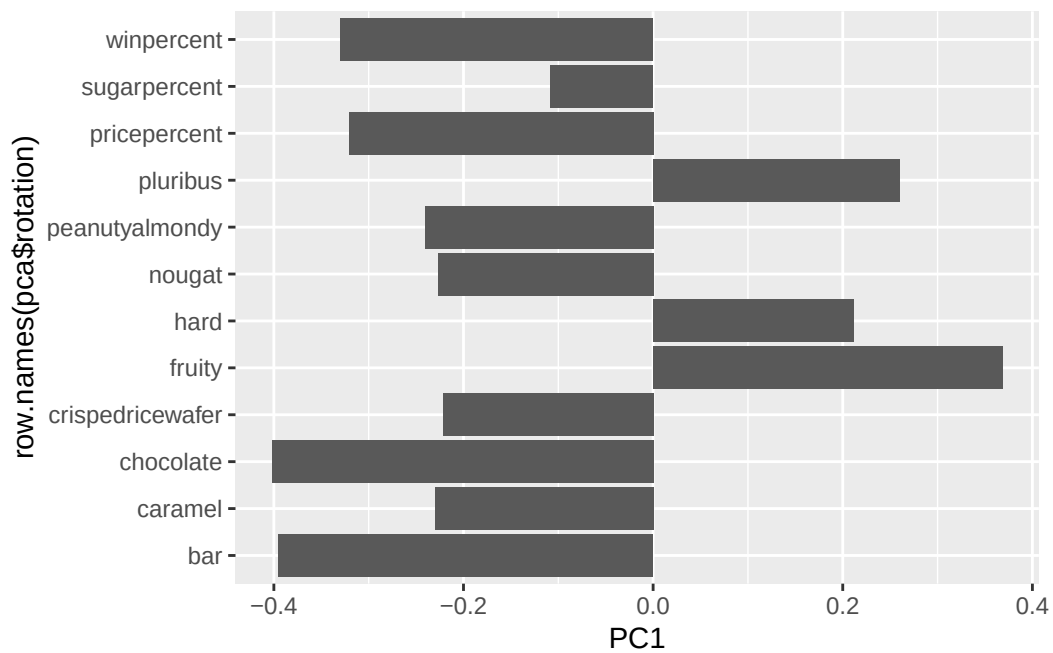
```
plot(pca$x[,1], pca$x[,2],  
     xlab = "PC1",  
     ylab = "PC2",  
     main = "PCA of Candy Dataset")
```



```
plot(pca$x[,1:2], col = my_cols, pch = 16)
```

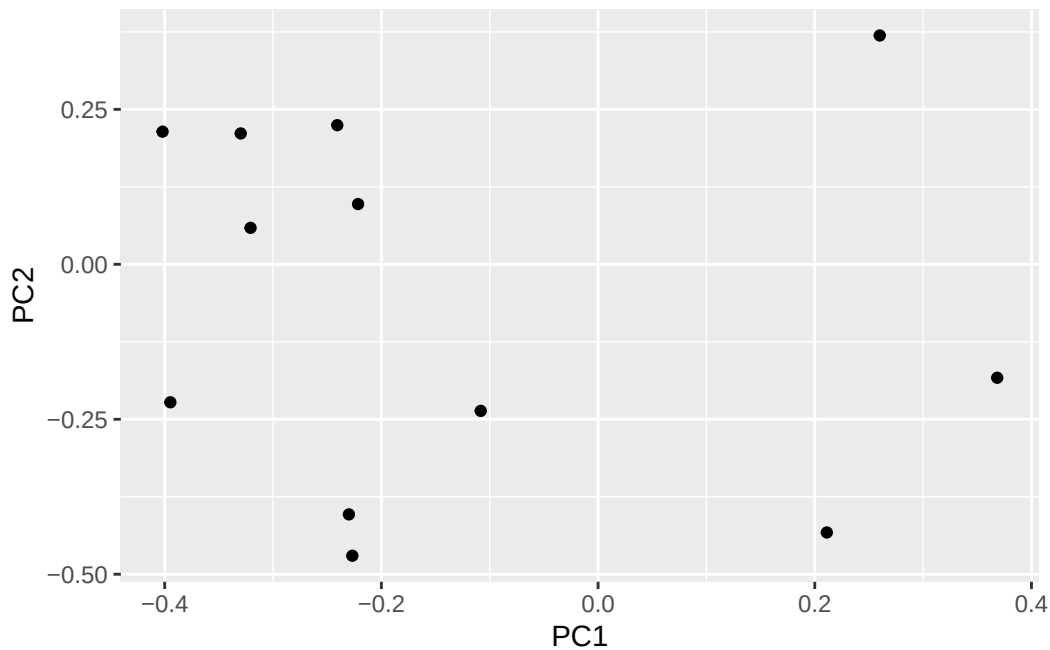


```
ggplot(pca$rotation) +
  aes(PC1, row.names(pca$rotation)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

```
ggplot(pca$rotation) +  
  aes(PC1, PC2) +  
  geom_point()
```



PC1 mainly separates chocolate-based candies from fruity/non-chocolate ones. Variables like chocolate, bar, peanuty, almondy, caramel, and nougat load strongly positive on PC1, while fruity, hard, and pluribus load strongly negative. This matches what we saw earlier in the correlation matrix and exploratory plots: chocolate candies cluster together and tend to be more similar to each other than to fruity candies.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Across the exploratory plots, correlation matrix, and PCA, the same pattern appears: the most successful candies are chocolate-based, often bars, and frequently include peanuts, almonds, caramel, or nougat. These traits cluster together in the correlation heatmap, load strongly in the positive direction on PC1, and dominate the high-winpercent end of the barplots. Fruity or hard candies consistently fall on the opposite side of PC1 and tend to have lower winpercent.

values. Together, these analyses show that chocolate + bar structure + nut/caramel components make a “winning” candy.